

Lecture: Gamma and Beta Distributions in Data Science

Suggested Teaching Flow

You can teach this in approximately **90–120 minutes**:

1. Quick recap: Discrete vs Continuous distributions
 2. Why Gamma Distribution?
 3. Gamma Distribution definition
 4. Gamma PDF
 5. Shape and Scale parameters
 6. Mean, variance, standard deviation
 7. Gamma solved problems
 8. Gamma vs Exponential vs Poisson
 9. Why Beta Distribution?
 10. Beta Distribution definition
 11. Beta PDF
 12. Shape parameters
 13. Mean and variance
 14. Beta solved problems
 15. Beta in A/B testing
 16. Beta as a Bayesian distribution
 17. Gamma as a Bayesian prior
 18. Gamma vs Beta — final comparison
 19. Short classroom exercises
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PART A — GAMMA DISTRIBUTION

1. Start With a Real-Life Question

Suppose a website receives customers continuously. How long do we have to wait until the **next customer** arrives?

This is modeled by the **Exponential Distribution**.

But now change the question:

How long do we have to wait until the **5th customer** arrives?

That's where the **Gamma Distribution** becomes useful.

Simple comparison

Exponential = time until 1 event
Gamma = time until several events

For example:

Question	Distribution
Time until next customer	Exponential
Time until 5 customers	Gamma
Number of customers in 10 minutes	Poisson

This is the first concept I would make students remember.

2. What Does Gamma Distribution Model?

The Gamma distribution is a **continuous probability distribution**.

It is commonly used to model:

The waiting time until several events occur.

If:

$X \sim \text{Gamma}(k, \theta)$

then:

- X = continuous random variable
- k = shape parameter
- θ = scale parameter

So:

$$X \sim \text{Gamma}(k, \theta)$$

3. Gamma PDF

The Gamma probability density function is:

$$f(x) = \frac{x^{k-1} e^{-x/\theta}}{\Gamma(k) \theta^k}$$

where:

$$x > 0$$

and:

- k = shape
- θ = scale
- $\Gamma(k)$ = Gamma function

4. What is the Gamma Function?

Students may ask:

"Sir/Madam, what is $\Gamma(k)$?"

Explain it simply.

For positive integers:

$$\Gamma(k) = (k - 1)!$$

For example:

$$\Gamma(1) = 0! = 1$$

$$\Gamma(2) = 1! = 1$$

$$\Gamma(3) = 2! = 2$$

$$\Gamma(4) = 3! = 6$$

$$\Gamma(5) = 4! = 24$$

So when solving numerical problems with integer k , students can generally replace:

$$\Gamma(k)$$

with:

$$(k-1)!$$

5. Shape Parameter k

The **shape parameter** controls the shape of the Gamma distribution.

Consider:

$\text{Gamma}(1, \theta)$

This is actually an **Exponential distribution**.

As k increases, the distribution changes shape.

Conceptually:

$k=1$

gives a strongly right-skewed distribution.

Larger k generally makes the distribution more concentrated and less skewed.

Important teaching point

Tell students:

k is related to the number of events we are waiting for.

For example:

$k=1$

→ waiting for 1 event

$k=3$

→ waiting for 3 events

$k=10$

→ waiting for 10 events

6. Scale Parameter θ

The scale parameter determines the **time scale**.

If:

θ

is large, waiting times tend to be larger.

If:

θ

is small, waiting times tend to be smaller.

7. Mean of Gamma Distribution

The mean is:

$$\mu = k\theta$$

This is one of the most important formulas.

Example

Suppose:

$$k=4$$

and:

$$\theta=3$$

Then:

$$\begin{aligned}\mu &= k\theta \\ \mu &= 4(3)\end{aligned}$$

$$\mu = 12$$

8. Variance

The variance is:

$$\sigma^2 = k\theta^2$$

For:

$$k=4, \quad \theta=3$$

we get:

$$\begin{aligned}\sigma^2 &= 4(3)^2 \\ \sigma^2 &= 4(9)\end{aligned}$$

$$\sigma^2 = 36$$

Therefore:

$$\sigma = \sqrt{36}$$

$$\sigma = 6$$

9. Important Gamma Formulas

Put these together on the board:

$$X \sim \text{Gamma}(k, \theta)$$

PDF

$$f(x) = \frac{x^{k-1}e^{-x/\theta}}{\Gamma(k)\theta^k}$$

Mean

$$\mu = k\theta$$

Variance

$$\sigma^2 = k\theta^2$$

Standard deviation

$$\sigma = \theta\sqrt{k}$$

For integer k

$$\Gamma(k) = (k-1)!$$

10. Gamma Solved Problem 1 — Website Visitors

Problem

A website receives an average of **2 visitors per minute**.

Find the expected time until the **5th visitor** arrives.

Step 1: Arrival rate

$$\lambda = 2 \text{ visitors/minute}$$

For an Exponential waiting time:

$$\theta = \frac{1}{\lambda}$$

Therefore:

$$\theta = \frac{1}{2} = 0.5$$

The 5th arrival means:

$$k=5$$

Therefore:

$$X \sim \text{Gamma}(5, 0.5)$$

Step 2: Mean

$$E(X) = k\theta$$

$$E(X) = 5(0.5)$$

$$E(X) = 2.5$$

Therefore:

$$E(X) = 2.5 \text{ minutes}$$

Interpretation

The expected time until the **5th visitor** is:

$$2.5 \text{ minutes}$$

11. Gamma Solved Problem 2 — API Requests

An API receives:

4 requests/second

Find the expected time until **10 requests** arrive.

Given

$$\lambda = 4$$

$$k = 10$$

Scale:

$$\theta = \frac{1}{4} = 0.25$$

Therefore:

$$X \sim \text{Gamma}(10, 0.25)$$

Mean:

$$E(X) = k\theta$$

$$E(X) = 10(0.25)$$

$$E(X) = 2.5 \text{ seconds}$$

Interpretation

On average, it takes:

2.5 seconds

for 10 API requests to arrive.

12. Gamma Solved Problem 3 — Mean, Variance

Suppose:

$$X \sim \text{Gamma}(3, 2)$$

Find:

1. Mean

2. Variance
3. Standard deviation

Mean

$$\mu = k\theta$$

$$\mu = 3(2)$$

$$\mu = 6$$

Variance

$$\sigma^2 = k\theta^2$$

$$\sigma^2 = 3(2)^2$$

$$\sigma^2 = 12$$

$$\sigma^2 = 12$$

Standard deviation

$$\sigma = \sqrt{12}$$

$$\sigma \approx 3.464$$

13. Gamma Solved Problem 4 — Probability

Suppose:

$$X \sim \text{Gamma}(3, 2)$$

Find:

$$P(X \leq 4)$$

For integer k , the Gamma CDF can be written as:

$$P(X \leq x) = 1 - e^{-x/\theta} \sum_{j=0}^{k-1} \frac{(x/\theta)^j}{j!}$$

Here:

$$k=3, \theta=2, x=4$$

Therefore:

$$\frac{x}{\theta} = \frac{4}{2} = 2$$

So:

$$\begin{aligned} P(X \leq 4) &= 1 - e^{-2} \left[1 + \frac{2}{1!} + \frac{2^2}{2!} \right] \\ P(X \leq 4) &= 1 - e^{-2}(1 + 2 + 2) \\ P(X \leq 4) &= 1 - 5e^{-2} \end{aligned}$$

Since:

$$e^{-2} \approx 0.1353$$

we get:

$$P(X \leq 4) = 1 - 5(0.1353)$$

$$P(X \leq 4) \approx 0.3233$$

Therefore:

$$32.33\%$$

14. Gamma and Exponential Connection

This is a **must-teach concept**.

Suppose the waiting times are:

$$T_1, T_2, T_3, \dots$$

and each waiting time is Exponential.

Then:

$$T_i \sim \text{Exponential}(\lambda)$$

The time until the k-th event is:

$$X = T_1 + T_2 + \dots + T_k$$

and:

$$X \sim \text{Gamma}(k, \theta)$$

where:

$$\theta = \frac{1}{\lambda}$$

Therefore:

Gamma = sum of several Exponential waiting times

That's the intuition behind Gamma.

PART B — BETA DISTRIBUTION

Now transition with this question:

"Gamma is useful for waiting time. But suppose I want to estimate the probability that a customer purchases a product. Which distribution should I use?"

The answer:

Beta Distribution

15. Why Do We Need Beta Distribution?

Suppose:

p = probability of customer purchase

A probability must satisfy:

$$0 \leq p \leq 1$$

We need a probability distribution whose values are restricted to:

[0,1]

The **Beta distribution** is ideal for this.

16. Definition of Beta Distribution

The Beta distribution is a **continuous probability distribution defined between 0 and 1**.

We write:

$$X \sim \text{Beta}(\alpha, \beta)$$

where:

- α = first shape parameter
- β = second shape parameter

Notice something important:

Gamma

$$\text{Gamma}(k, \theta)$$

uses:

- shape
- scale

Beta

$$\text{Beta}(\alpha, \beta)$$

uses:

- shape
- shape

17. Beta PDF

The PDF is:

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$$

where:

$$0 < x < 1$$

and:

$$B(\alpha, \beta)$$

is called the **Beta function**.

18. What is Beta Function?

The Beta function is:

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

For positive integers, Gamma functions can be simplified using factorials.

For example:

$$\begin{aligned} B(2,3) &= \frac{\Gamma(2)\Gamma(3)}{\Gamma(5)} \\ B(2,3) &= \frac{1! \, 2!}{4!} \\ B(2,3) &= \frac{2}{24} \end{aligned}$$

$$B(2,3) = \frac{1}{12}$$

19. Mean of Beta Distribution

The mean is:

$$E(X) = \frac{\alpha}{\alpha + \beta}$$

This is very important.

Example

If:

$$X \sim \text{Beta}(2,3)$$

then:

$$E(X) = \frac{2}{2+3}$$

$$E(X) = 0.4$$

So the distribution is centered around 0.4.

20. Variance of Beta Distribution

The variance is:

$$\text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

For:

Beta(2,3)

we have:

$$\begin{aligned}\text{Var}(X) &= \frac{(2)(3)}{(5)^2(6)} \\ \text{Var}(X) &= \frac{6}{150}\end{aligned}$$

$$\text{Var}(X) = 0.04$$

21. Beta Solved Problem 1 — Website Conversion Rate

Suppose a Data Scientist wants to estimate the probability that a website visitor purchases a product.

Let:

p = purchase probability

Suppose:

$p \sim \text{Beta}(2,3)$

Find the expected purchase probability.

Solution

Using:

$$E(p) = \frac{\alpha}{\alpha + \beta}$$

we get:

$$E(p) = \frac{2}{2+3}$$

$$E(p) = \frac{2}{5}$$

$$E(p) = 0.4$$

Therefore:

40%

Interpretation

Before observing additional data, the expected conversion probability is **40%** according to this Beta model.

22. Beta Solved Problem 2 — A/B Testing

Suppose an e-commerce website has a conversion rate p .

Initially, the Data Scientist assumes:

$$p \sim \text{Beta}(2,2)$$

This represents a prior belief centered at:

$$E(p) = \frac{2}{2+2}$$

$$E(p) = 0.5$$

So initially, the expected conversion probability is:

50%

Now collect data

Suppose:

- 10 customers visit
- 7 purchase
- 3 do not purchase

So:

successes=7

failures=3

For a Beta-Binomial model:

$$Beta(\alpha, \beta) + \text{data} \rightarrow Beta(\alpha + s, \beta + f)$$

Therefore:

Beta(2,2)

becomes:

Beta(2+7,2+3)

$$Beta(9,5)$$

Posterior Mean

$$E(p \mid \text{Data}) = \frac{9}{9+5}$$

$$E(p \mid \text{Data}) = \frac{9}{14}$$

$$E(p \mid \text{Data}) \approx 0.643$$

Therefore:

$$64.3\%$$

Interpretation

After observing the data, our updated estimate of the conversion probability is approximately:

$$64.3\%$$

23. Why Not Just Say 70%?

Students will probably ask:

"But 7 out of 10 purchased, so why aren't we saying 70%?"

Excellent question.

The observed proportion is:

$$\frac{7}{10} = 0.7$$

or:

70%

But Bayesian analysis combines:

Prior information + Observed data

The prior was:

Beta(2,2)

After seeing the data:

Beta(9,5)

So the posterior estimate is:

64.3%

The difference is because we are incorporating **prior uncertainty**.

With a very large amount of data, the influence of a reasonable prior generally becomes smaller.

24. Beta Solved Problem 3 — Email Click Probability

An organization wants to estimate the probability that a customer clicks an email.

Suppose:

$p \sim \text{Beta}(1,4)$

Find the expected click probability.

Solution

$$E(p) = \frac{\alpha}{\alpha + \beta}$$

$$E(p) = \frac{1}{1+4}$$

$$E(p) = \frac{1}{5}$$

Therefore:

$$E(p) = 0.20$$

or:

$$20\%$$

Interpretation

The prior expected email click probability is **20%**.

25. Beta Solved Problem 4 — Variance

Suppose:

$$X \sim \text{Beta}(2,3)$$

Find the variance.

Using:

$$\text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

Substitute:

$$\text{Var}(X) = \frac{(2)(3)}{(2+3)^2(2+3+1)}$$

$$\text{Var}(X) = \frac{6}{25(6)}$$

$$\text{Var}(X) = \frac{6}{150}$$

Therefore:

$$\text{Var}(X) = 0.04$$

26. How Beta Shape Parameters Work

This is worth explaining visually on the board.

Consider:

Beta(1,1)

This gives the **Uniform distribution from 0 to 1**.

That's a beautiful connection:

$$Beta(1,1) = Uniform(0,1)$$

Now consider:

Beta(2,2)

This tends to concentrate around:

0.5

Whereas:

Beta(5,2)

places more weight toward higher probabilities.

And:

Beta(2,5)

places more weight toward lower probabilities.

Intuition

α larger relative to $\beta \Rightarrow$ probability tends toward 1
 β larger relative to $\alpha \Rightarrow$ probability tends toward 0

27. The Most Important Data Science Application — A/B Testing

Imagine you have two website designs.

Website A

100 visitors:

12 purchases

Conversion:

$$\frac{12}{100} = 0.12$$

Website B

100 visitors:

18 purchases

Conversion:

$$\frac{18}{100} = 0.18$$

A simple analysis says:

$A = 12\%$

$B = 18\%$

But a Bayesian Data Scientist can model:

$$p_A \sim \text{Beta}(\alpha_A, \beta_A)$$

and:

$$p_B \sim \text{Beta}(\alpha_B, \beta_B)$$

Then after observing the data, we obtain:

$$p_A \mid \text{Data}$$

and:

$$p_B \mid \text{Data}$$

and ask:

$$P(p_B > p_A \mid \text{Data})$$

This answers the practical question:

What is the probability that B is actually better than A?

That's why Beta is extremely useful in **A/B testing and conversion-rate analysis**.

PART C — GAMMA IN BAYESIAN DATA SCIENCE

Now return to Gamma.

Ask students:

"Beta models a probability. What if I want to model a rate?"

For example:

λ = website requests per minute

A rate must be:

$$\lambda > 0$$

Gamma is suitable:

$$\lambda \sim \text{Gamma}(\alpha, \beta)$$

This is a **Bayesian prior** for the rate.

28. Gamma as a Bayesian Prior

Suppose:

λ = number of customer arrivals per minute

We choose:

$$\lambda \sim \text{Gamma}(2, 1)$$

The Gamma distribution describes our uncertainty about the unknown rate.

Then we collect data.

Suppose:

30 customers

arrive during:

10 minutes

The observed rate is:

$$\frac{30}{10} = 3$$

The Bayesian model combines:

Prior + Data $\rightarrow$ Posterior

For a Poisson likelihood and Gamma prior, the posterior is also Gamma.

This is called a:

conjugate prior

29. Beta-Binomial and Gamma-Poisson

This is perhaps the **single most important connection** for your students.

Case 1: Binary data

If the data is:

Success/Failure

and:

$X \sim \text{Binomial}(n, p)$

then:

$$p \sim \text{Beta}$$

and:

$$\text{Beta} + \text{Binomial} \rightarrow \text{Beta}$$

Case 2: Count data

If the data is:

$0, 1, 2, 3, \dots$

and:

$X \sim \text{Poisson}(\lambda)$

then:

$$\lambda \sim \text{Gamma}$$

and:

$$\text{Gamma} + \text{Poisson} \rightarrow \text{Gamma}$$

30. The Big Picture

Write this on the board:

$$\text{Binomial} \rightarrow \text{probability } p \rightarrow \text{Beta}$$

and:

$$\text{Poisson} \rightarrow \text{rate } \lambda \rightarrow \text{Gamma}$$

This makes the entire topic much easier.

31. Beta vs Gamma — Clear Comparison

Feature	Beta	Gamma
Distribution type	Continuous	Continuous
Values	$0 < x < 1$	$x > 0$
Parameters	α, β	k, θ
Parameters are	Shape, Shape	Shape, Scale
Models	Probability/proportion	Positive values/rates/waiting times
Mean	$\frac{\alpha}{\alpha + \beta}$	$k\theta$
Variance	$\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$	$k\theta^2$

Feature	Beta	Gamma
Example	Conversion probability	Time until 5 arrivals
Bayesian use	Prior for probability	Prior for rate
Common likelihood	Binomial	Poisson

32. Very Important Terminology

Don't tell students:

"Gamma is a hyperparameter."

Instead explain:

Gamma distribution

is a **probability distribution**.

Gamma parameters

k, θ

are the parameters of that distribution.

In Bayesian modeling

If Gamma is used as a prior:

$\lambda \sim \text{Gamma}(\alpha, \beta)$

then α, β are hyperparameters of the prior distribution.

Similarly:

$p \sim \text{Beta}(\alpha, \beta)$

has α, β as hyperparameters of the Beta prior.

33. Common Student Confusion — Clear This Up

Gamma Distribution vs Gamma Function

These are different.

Gamma distribution

$\text{Gamma}(k, \theta)$

is a probability distribution.

Gamma function

$\Gamma(k)$

is a mathematical function used in the PDF.

Don't confuse:

Gamma distribution

with:

$\Gamma(k)$ *Gamma function*

34. Another Important Confusion — Gamma Scale vs Rate

Your slide uses:

$\text{Gamma}(k, \theta)$

where θ is the scale parameter.

Therefore:

$$E(X) = k\theta$$

and:

$$\text{Var}(X) = k\theta^2$$

But Bayesian textbooks sometimes write:

$\text{Gamma}(\alpha, \beta)$

where β is a rate parameter.

Then:

$$E(X) = \frac{\alpha}{\beta}$$

So tell students:

Always check whether Gamma is being parameterized using scale or rate.

This prevents a LOT of numerical mistakes.

35. Quick Classroom Questions

After explaining the concepts, give these without solutions first.

Question 1

A website receives 5 visitors per minute.

What distribution models the time until the 4th visitor?

Answer:

$$\text{Gamma}(4, 0.2)$$

because:

$$\theta = \frac{1}{5} = 0.2$$

Question 2

If:

$$X \sim \text{Gamma}(4, 3)$$

find the mean.

$$E(X) = k\theta$$

$$E(X) = 4(3)$$

$$12$$

Question 3

If:

$X \sim \text{Beta}(3,7)$

find the mean.

$$E(X) = \frac{3}{3+7}$$

0.3

Question 4

Which distribution is suitable for modeling customer conversion probability?

Beta

Question 5

Which distribution is suitable for modeling the number of server requests per second?

For the **count of requests in an interval**:

Poisson

If modeling the **unknown positive request rate** in Bayesian analysis:

Gamma

36. Final "Story" to Give Students

I'd finish the lecture with this simple story:

Imagine an online shopping website.

Question 1

How many customers visit in 10 minutes?

Use:

Poisson

Question 2

How long until the next customer arrives?

Use:

Exponential

Question 3

How long until the 5th customer arrives?

Use:

Gamma

Question 4

What is the probability that a customer purchases something?

Use:

Beta

Question 5

What is the unknown customer-arrival rate?

In Bayesian modeling, use:

Gamma

Question 6

What is the probability that Website B has a higher conversion rate than Website A?

Use:

Beta distributions

for the two conversion probabilities.

One-Page Board Summary

At the end of the class, write this:

Gamma

$X \sim \text{Gamma}(k, \theta)$

Use: waiting time until several events / positive quantities.

$$f(x) = \frac{x^{k-1}e^{-x/\theta}}{\Gamma(k)\theta^k}$$
$$\mu = k\theta$$
$$\sigma^2 = k\theta^2$$
$$\sigma = \theta\sqrt{k}$$

Example:

Time until 5 customers arrive.

Beta

$X \sim \text{Beta}(\alpha, \beta)$

Use: probabilities and proportions.

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$$
$$\mu = \frac{\alpha}{\alpha + \beta}$$
$$\text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

Example:

Probability that a customer purchases a product.



The 6-Line Memory Trick

Tell your students to remember just this:

Bernoulli → 1 trial
Binomial → number of successes
Geometric → trials until 1st success
Poisson → number of events
Exponential → waiting time for 1 event
Gamma → waiting time for k events

And for the Bayesian side:

Beta → probability p Gamma → positive rate λ